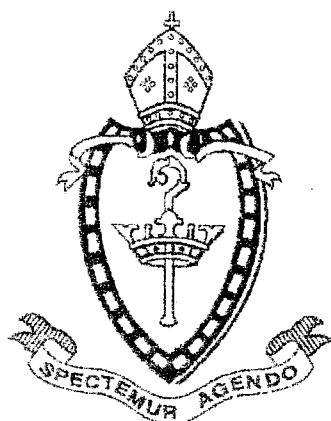


NEWCASTLE GRAMMAR SCHOOL



YEAR 12

2004

EXTENSION 2 MATHEMATICS TRIAL EXAMINATION

*Time allowed – Three hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on page 12.
- Board-approved calculators may be used.
- Answer each question in a SEPARATE Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

QUESTION 1

Use a SEPARATE Writing Booklet

Marks

a) Evaluate $\int_0^1 te^{-t} dt$ 3

b) i) Find the real numbers a , b and c such that 2

$$\frac{1}{x(1+x^2)} \equiv \frac{a}{x} + \frac{bx+c}{1+x^2}$$

ii) Hence find $\int \frac{dx}{x(1+x^2)}$ 2

c) Evaluate $\int_0^4 \frac{x}{\sqrt{x+4}} dx$ 3

d) i) If $I_n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$ show that, for $n > 1$ 3

$$I_n = \left(\frac{\pi}{2}\right)^n - n(n-1)I_{n-2}$$

ii) Hence find the area of the region bounded by 2
the curve $y = x^4 \cos x$ and the x -axis for $0 \leq x \leq \frac{\pi}{2}$

QUESTION 2

Use a SEPARATE Writing Booklet

Marks

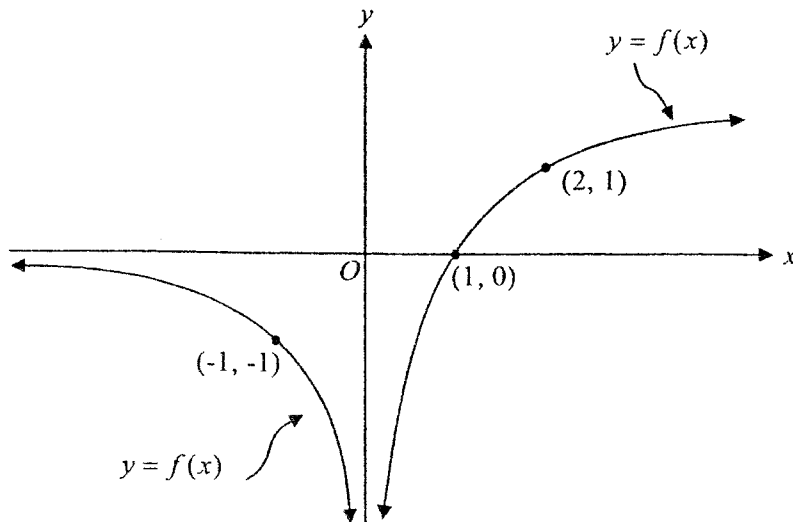
- a) The complex number z moves such that $\operatorname{Im}\left(\frac{1}{z-i}\right) = 1$. 3
 Show that the locus of z is a circle and find its centre and radius.
- b) i) Find the square roots of the complex number $5 - 12i$ 2
- ii) Given that $z = \frac{1 + \sqrt{5 - 12i}}{2 + 2i}$ and is purely imaginary, 2
 find z^{400}
- c) i) Shade the region on the Argand diagram containing all of the points representing the complex numbers z such that 3
 $|z - 1 - i| \leq 1$ and $-\frac{\pi}{4} \leq \arg(z - i) \leq \frac{\pi}{4}$
- ii) Let w be the complex number of minimum modulus satisfying the inequalities of part i) above. 1
 Express w in the form $x + iy$.
- d) Express $z = \frac{-1 + i}{\sqrt{3} + i}$ in modulus/argument form and hence 4
 evaluate $\cos \frac{7\pi}{12}$ in surd form.

QUESTION 3

Use a SEPARATE Writing Booklet

Marks

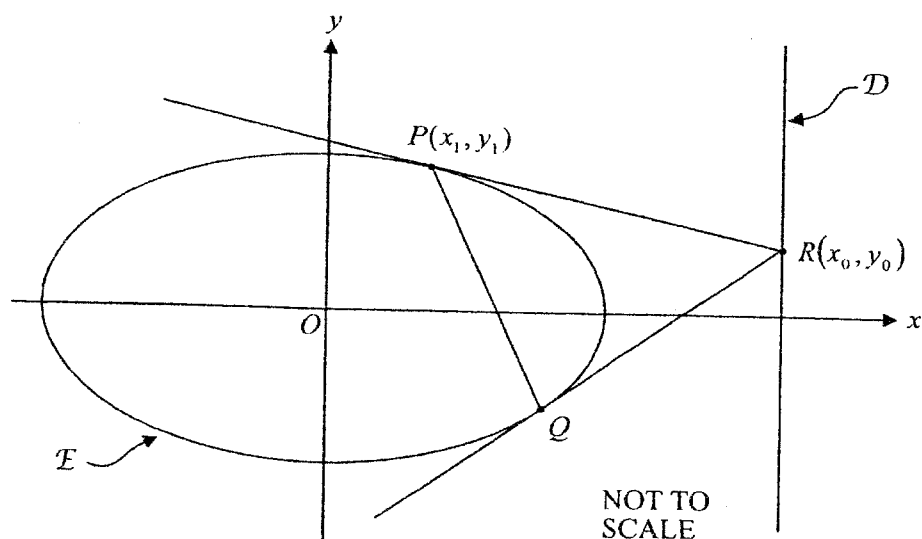
- a) The diagram below shows the graph of the discontinuous function $y = f(x)$



Draw large (half page), separate sketches of the following

- | | | |
|------|----------------------|---|
| i) | $y = -\sqrt{f(x)}$ | 3 |
| ii) | $y = f(x) $ | 3 |
| iii) | $y = \frac{1}{f(x)}$ | 3 |

b)



The ellipse $\mathcal{E}$ with equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$ has a directrix $\mathcal{D}$ as shown in the diagram. Point $R(x_0, y_0)$ lies on $\mathcal{D}$. PQ is the chord of contact from R where P is the point (x_1, y_1) .

i) Write down the equation of $\mathcal{D}$ 1

ii) Show that the equation of the tangent at P is 3

$$\frac{x_1 x}{25} + \frac{y_1 y}{16} = 1$$

iii) The equation of PQ is $\frac{x_0 x}{25} + \frac{y_0 y}{16} = 1$ 2

Show that the focus of $\mathcal{E}$ lies on PQ

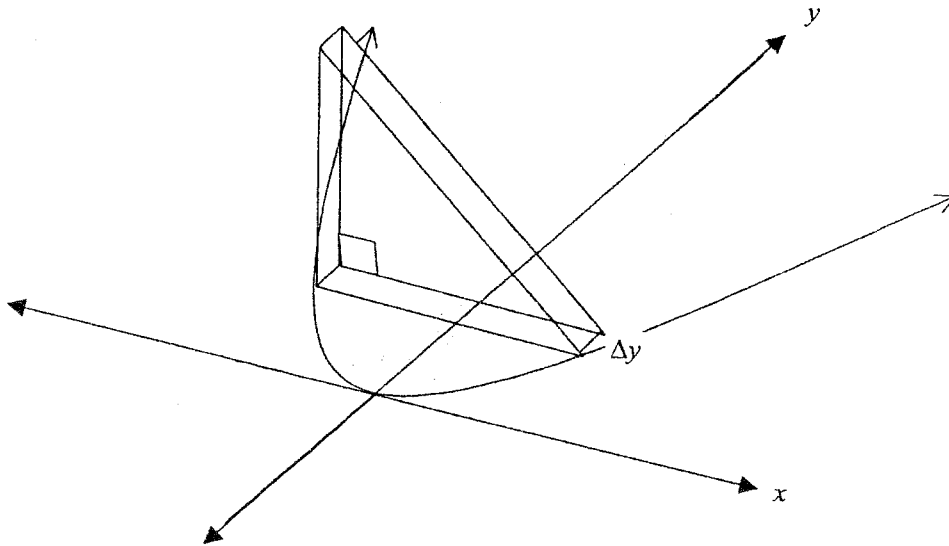
QUESTION 4

Use a SEPARATE Writing Booklet

Marks

- a) A solid is formed as shown below. Its base is in the xy -plane and is in the shape of the parabola $y = x^2$. The vertical cross-section is in the shape of a right angled isosceles triangle.

4

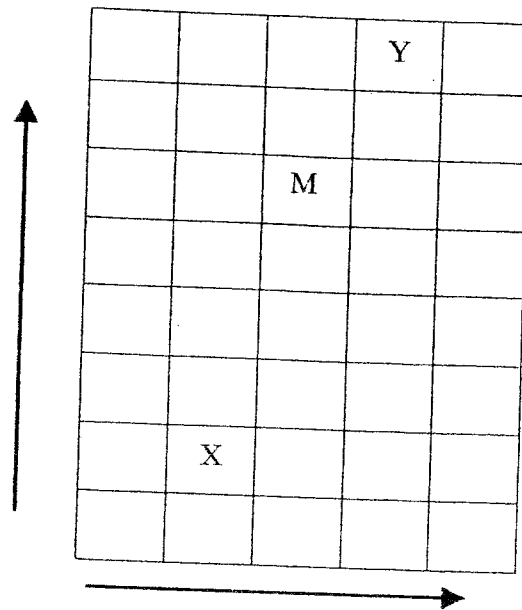


By using the method of slicing, calculate the volume of the solid between the values $y = 0$ and $y = 4$.

- b) Find, using the method of cylindrical shells, the volume of the solid generated by rotating the region bounded by the curve $y = (x - 2)^2$ and the line $y = x$ about the x -axis.

6

- c) On a special chess board, the squares are arranged in 8 rows and 5 columns as shown



A player can only move forwards or across in the directions shown by the arrows, one square at a time.

- i) If a player is situated at X, in how many ways can the player reach the square labelled Y? 3
- ii) In how many ways can a player move from X to Y if they must pass through M? 2

QUESTION 5

Use a SEPARATE Writing Booklet

Marks

a) The cubic equation $x^3 - x^2 + 4x - 2 = 0$ has roots α, β and γ

i) Find the equation with the roots α^2, β^2 and γ^2 3

ii) Find the value of $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$ 3

b) If $P(x) = 4x^3 + 4x^2 + x + k$ for some real number k , find the values of x for which $P'(x) = 0$. Hence find the values of k for which the equation $P(x) = 0$ has more than one real root. 4

c) If $P(x) = 3x^4 - 11x^3 + 14x^2 - 11x + 3$ show that 5

$$P(x) = x^2 \left\{ 3 \left(x + \frac{1}{x} \right)^2 - 11 \left(x + \frac{1}{x} \right) + 8 \right\}$$

and hence solve $P(x) = 0$ over C (complex numbers)
and factorise $P(x)$ over R (real numbers)

QUESTION 6

Use a SEPARATE Writing Booklet

Marks

- a) i) Show that the equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$ is

4

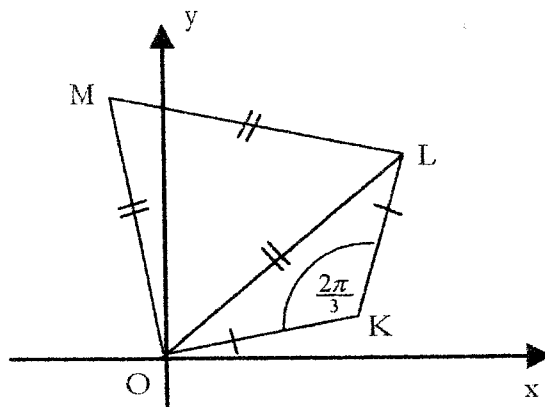
$$a \sin \theta x + by = (a^2 + b^2) \tan \theta$$

- ii) The normal at the point $P(a \sec \theta, b \tan \theta)$ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the x -axis at G . PN is the perpendicular from P to the x -axis. Prove that $OG = e^2 \times ON$, where O is the origin.

5

- b) The points K and M in a complex plane represent the complex numbers α and β respectively. The triangle OKL is isosceles and $\angle OKL = \frac{2\pi}{3}$. The triangle OLM is equilateral. Show that $3\alpha^2 + \beta^2 = 0$

6



QUESTION 7

Use a SEPARATE Writing Booklet

Marks

- a) Prove by induction that, for $n \geq 1$

5

$$\cos \frac{90^\circ}{2^n} = \frac{1}{2} \underbrace{\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}}_{n\text{-terms}}$$

- b) i) Prove that:

3

$$\tan^{-1}(n+1) - \tan^{-1}(n) = \cot^{-1}(1+n+n^2)$$

- ii) Hence, sum the series

3

$$\cot^{-1} 3 + \cot^{-1} 7 + \cot^{-1} 13 + \dots + \cot^{-1}(1+n+n^2)$$

- c) Using a graph, find the values of x for which $f(x) > (f(x))^3$
where $f(x) = \frac{1}{2} + \sin x$ and $0 \leq x \leq 2\pi$

4

QUESTION 8

Use a SEPARATE Writing Booklet

Marks

- a) The tangent at $P(cp, \frac{c}{p})$ to the hyperbola $xy = c^2$ meets the lines $y = \pm x$ at A and B respectively. The normal at P meets the axes at C and D . If M represents the area of $\triangle OAB$ and N represents the area of $\triangle OCD$ show that M^2N is a constant. 6
- b) i) Determine whether $f(x) = \frac{1-|x|}{|x|}$ is even, odd or neither. 1
Justify your answer.
- ii) Sketch $y = f(x)$ 3
- iii) Hence, or otherwise, solve $f(x) \geq 1$ 3
- iv) Sketch $y = e^{f(x)}$ 2

$$(a) \int_0^1 t e^{-t} dt$$

$$\text{let } u=t \quad v'=e^{-t}$$

$$\therefore u'=1 \quad \therefore v=-e^{-t}$$

$$[-t e^{-t}]_0^1 + \int_0^1 e^{-t} dt$$

$$[-e^{-t}-0] + -[e^{-t}]_0^1$$

$$-e^{-1} - (e^{-1}-1)$$

$$\boxed{1-2e^{-1} \text{ or } 1-\frac{2}{e}}$$

(3)

$$\frac{1}{x(1+x^2)} \equiv \frac{a}{x} + \frac{bx+c}{1+x^2} \quad \text{x B.S.}$$

$$1 = a(1+x^2) + (bx+c)x$$

$$= ax^2 + a + bx^2 + cx$$

$$0x^2 + 0x + 1 = (a+b)x^2 + cx + a$$

equating coefficients:

$$a+b=0$$

$$c=0$$

$$a=1$$

$$\Rightarrow \boxed{a=1, b=-1, c=0}$$

(2)

$$\int \frac{dx}{x(1+x^2)} = \int \frac{1}{x} - \frac{x}{1+x^2} dx \quad \text{from (i)}$$

$$= \int \frac{1}{x} - \frac{1}{2} \cdot \frac{2x}{1+x^2} dx$$

$$= \boxed{\ln|x| - \frac{1}{2} \ln|1+x^2| + c} \quad (2)$$

$$\int_0^4 \frac{x}{\sqrt{x+4}} dx$$

$$\text{let } u=x+4$$

$$\therefore du=dx$$

$$\int \frac{u-4}{\sqrt{u}} du$$

$$x=0 \Rightarrow u=4$$

$$x=4 \Rightarrow u=8$$

$$\int \frac{u}{\sqrt{u}} - \frac{4}{\sqrt{u}} du$$

$$\int u^{\frac{1}{2}} - 4u^{-\frac{1}{2}} du$$

$$[\frac{2}{3}u^{\frac{3}{2}} - 8u^{\frac{1}{2}}]_4^8$$

$$= (\frac{2}{3}(8)^{\frac{3}{2}} - 8(2\sqrt{2})) - (\frac{2}{3}(2)^{\frac{3}{2}} - 8(2))$$

$$= \frac{32\sqrt{2}}{3} - 16\sqrt{2} - \frac{16}{3} + 16$$

$$= \boxed{\frac{32-16\sqrt{2}}{3} \text{ or } \frac{16}{3}(2-\sqrt{2})} \quad (3)$$

(d)

$$(i) \text{ For: } I_n = \int_0^{\frac{\pi}{2}} x^n \cos x dx$$

$$\text{let } u=x^n \quad v'=\cos x$$

$$\therefore u'=nx^{n-1} \quad \therefore v=\sin x$$

$$\therefore I_n = [x^n \sin x]_0^{\frac{\pi}{2}} - n \int_0^{\frac{\pi}{2}} x^{n-1} \sin x dx$$

$$\text{let } u=x^{n-1} \quad v'=\sin x$$

$$\therefore u'=(n-1)x^{n-2} \quad \therefore v=-\cos x$$

$$\therefore I_n = \left(\frac{\pi}{2}\right)^n - n \left\{ [-x^{n-1} \cos x]_0^{\frac{\pi}{2}} + (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \cos x dx \right\}$$

$$= \left(\frac{\pi}{2}\right)^n - n[0] - n(n-1) I_{n-2}$$

$$\text{i.e. } I_n = \left(\frac{\pi}{2}\right)^n - n(n-1) I_{n-2} \quad (\text{QED}) \quad (3)$$

$$(ii) \text{ Area} = \int_0^{\frac{\pi}{2}} x^4 \cos x dx = I_4$$

(above x-axis for $0 \leq x \leq \frac{\pi}{2}$ as x^4 and $\cos x$ both ≥ 0 for $0 \leq x \leq \frac{\pi}{2}$)

$$I_4 = \left(\frac{\pi}{2}\right)^4 - 4(3) I_2$$

$$I_2 = \left(\frac{\pi}{2}\right)^2 - 2(1) I_0$$

$$\text{and } I_0 = \int_0^{\frac{\pi}{2}} x^0 \cos x dx$$

$$= \int_0^{\frac{\pi}{2}} dx$$

$$= [x]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2}$$

$$\therefore I_4 = \left(\frac{\pi}{2}\right)^4 - 12 \left\{ \left(\frac{\pi}{2}\right)^2 - 2\left(\frac{\pi}{2}\right) \right\}$$

$$= \boxed{\frac{\pi^4}{16} - 3\pi^2 + 12\pi \text{ units}^2} \quad (3)$$

a) Let $z = x + iy$

$\therefore \bar{z} = x - iy$

$\therefore \bar{z} - i = x - i(y+1)$

$\therefore \frac{1}{\bar{z} - i} = \frac{1}{x - i(y+1)} \times \frac{x + i(y+1)}{x + i(y+1)}$

$= \frac{x + i(y+1)}{x^2 + (y+1)^2}$

and we are given:

$\text{Im}\left(\frac{1}{\bar{z} - i}\right) = \frac{y+1}{x^2 + (y+1)^2} = 1$

i.e. $y+1 = x^2 + (y+1)^2$

$\therefore x^2 + y^2 + y = 0$

$x^2 + y^2 + y + \frac{1}{4} = \frac{1}{4}$

$x^2 + (y + \frac{1}{2})^2 = \frac{1}{4}$ (3)

$\therefore$ locus of z is circle, centre $(0, -\frac{1}{2})$
radius = $\frac{1}{2}$ unit

b)

(i) let $x + iy = \sqrt{5 - 12i}$

$\therefore (x + iy)^2 = 5 - 12i$

i.e. $x^2 - y^2 + 2ixy = 5 - 12i$

equating parts:

$x^2 - y^2 = 5 \dots (1) \quad 2xy = -12$

or $x = -6/y \dots (2)$

(2) into (1): $\frac{36}{y^2} - y^2 = 5$

$\therefore y^4 + 5y^2 - 36 = 0$

$(y^2 + 9)(y^2 - 4) = 0$

giving $y^2 = 4$ or $y = \pm 2$

NOTE: and $x = \mp 3$

$y^2 = -9$: y must be real.

$\therefore \sqrt{5 - 12i} = 3 - 2i$ or $-3 + 2i$ (2)

(ii) $z = \frac{1 + \sqrt{5 - 12i}}{2 + 2i}$

$\therefore z = \frac{1 + 3 - 2i}{2 + 2i}$ or $z = \frac{1 - 3 + 2i}{2 + 2i}$

$= \frac{4 - 2i}{2 + 2i} = \frac{-2 + 2i}{2 + 2i}$

$= \frac{2 - i}{1 + i} \times \frac{1 - i}{1 - i} = \frac{-1 + i}{1 + i} \times \frac{1 - i}{1 - i}$

$= \frac{1 - 3i}{2} = i$

choose $z = i$ (for z purely imaginary)

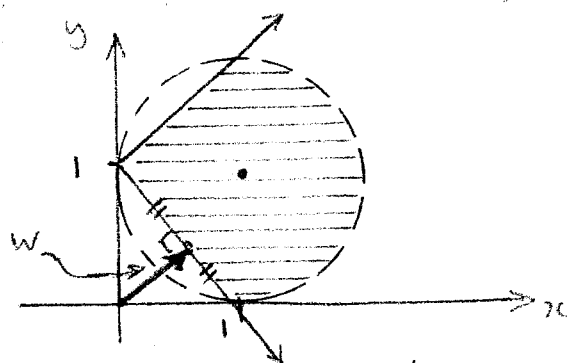
$\therefore z^{400} = i^{400}$
 $= (i^4)^{100}$
 $= 1^{100}$
 $= 1$ (2)

c)

(i) $|z - 1 - i| \leq 1 \equiv |z - (1 + i)| \leq 1$

i.e. inside circle, centre $1 + i$, radius = 1

$-\frac{\pi}{4} \leq \arg(z - i) \leq \frac{\pi}{4} \equiv$ angle from i
between $-\frac{\pi}{4}$ and $\frac{\pi}{4}$



(ii) w has minimum modulus

$\therefore$ shortest distance to line as indicated by w on diagram above.

i.e. to midpoint of $(0, 1)$ and $(1, 0)$ } $= (\frac{1}{2}, \frac{1}{2})$

$\therefore w = \frac{1}{2} + \frac{1}{2}i$ (1)

$$1) z = \frac{-1+i}{\sqrt{3}+i}$$

$$= \frac{\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}{2 \operatorname{cis} \frac{\pi}{6}}$$

$$= \frac{\sqrt{2}}{2} \operatorname{cis} \left(\frac{3\pi}{4} - \frac{\pi}{6} \right)$$

$$\therefore z = \frac{1}{\sqrt{2}} \operatorname{cis} \frac{7\pi}{12} \dots (1)$$

AND $z = \frac{-1+i}{\sqrt{3}+i} \times \frac{\sqrt{3}-i}{\sqrt{3}-i}$

$$= \frac{1-\sqrt{3} + (1+\sqrt{3})i}{4} \dots (2)$$

Equating real parts in (1) and (2):

$$\frac{1}{\sqrt{2}} \cos \frac{7\pi}{12} = \frac{1-\sqrt{3}}{4}$$

$$\therefore \cos \frac{7\pi}{12} = \frac{\sqrt{2}-\sqrt{6}}{4} \quad (4)$$

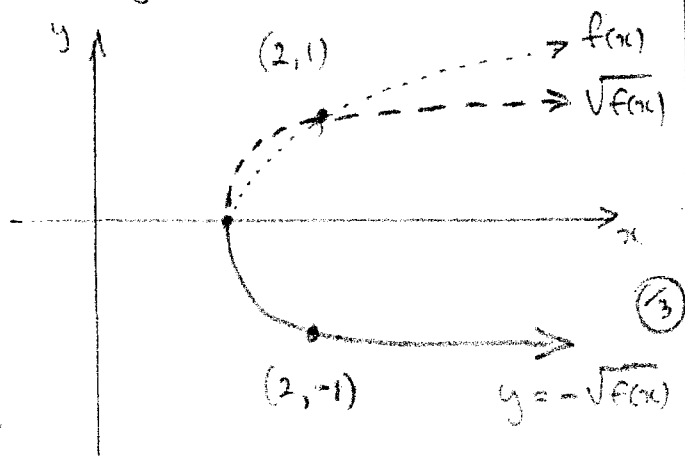
(a) (i) $y = \sqrt{f(x)}$ only defined
if $f(x) \geq 0$ (i.e. above/on x-axis)

• $y = \sqrt{f(x)}$ above $y = f(x)$

• $0 < f(x) < 1$, through $f(x) = 1$,

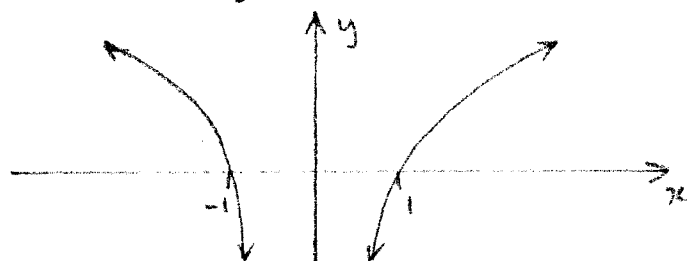
below $y = f(x)$ for $f(x) > 1$

• $y = -\sqrt{f(x)}$ below x-axis.

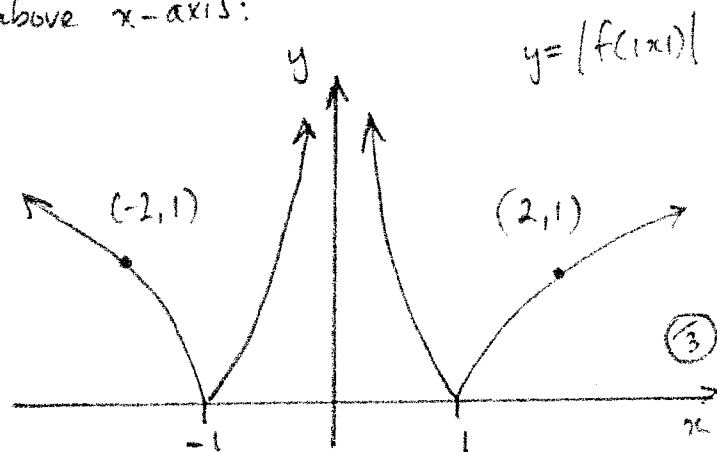


Question did not instruct you to sketch
• $y = f(x)$ or $y = \sqrt{f(x)}$

(ii) $y = f(|x|)$ has right side
reflected in y-axis.



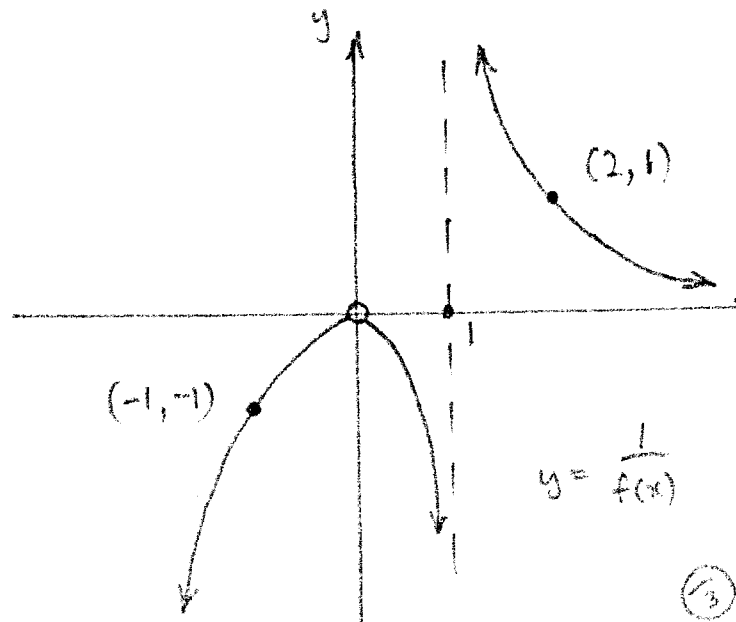
• $y = |f(|x|)|$ has those parts of
above graph below x-axis, reflected
above x-axis:



(iii) • $f(x) = 0$ at $x = 1 \Rightarrow$ asymptote
• $x \rightarrow 0, f(x) \rightarrow -\infty \Rightarrow \frac{1}{f(x)} \rightarrow 0$ from
below
(but undefined)

• $x \rightarrow -\infty, f(x) \rightarrow 0^- \Rightarrow \frac{1}{f(x)} \rightarrow -\infty$

• $x \rightarrow \infty, f(x) \rightarrow \infty \Rightarrow \frac{1}{f(x)} \rightarrow 0^+$



b) (i) D has equation: $x = \frac{a}{e}$

and for $b^2 = a^2(1 - e^2)$

$$16 = 25(1 - e^2)$$

$$\therefore e^2 = 1 - \frac{16}{25}$$

$$\therefore e = \frac{3}{5} \quad (e > 0)$$

$$\therefore x = \frac{5}{\frac{3}{5}} \quad (\text{for } a^2 = 25)$$

$$\therefore \boxed{x = \frac{25}{3}} \quad (1)$$

ii) Equation of tangent:

$$y - y_1 = m(x - x_1)$$

for m: $\frac{x^2}{25} + \frac{y^2}{16} = 1$ } implicit diff'n

$$\therefore \frac{2x}{25} + \frac{2y}{16} \cdot y' = 0$$

$$\therefore y' = -\frac{2x}{25} \times \frac{8}{y}$$

$$= -\frac{16x}{25y}$$

$$\therefore \text{at } P(x_1, y_1), m = -\frac{16x_1}{25y_1}$$

$\therefore$ equation is:

$$y - y_1 = -\frac{16x_1}{25y_1}(x - x_1)$$

$$\therefore 25y_1y - 25y_1^2 = -16x_1x + 16x_1^2$$

$$\therefore 16x_1x + 25y_1y = 16x_1^2 + 25y_1^2$$

($\div 16 \times 25$)

$$\therefore \frac{x_1x}{25} + \frac{y_1y}{16} = \frac{x_1^2}{25} + \frac{y_1^2}{16} = 1$$

$$\therefore \frac{x_1x}{25} + \frac{y_1y}{16} = 1 \quad (QED) \quad (3)$$

(iii) Focus at $(ae, 0)$

i.e. at $(5 \times \frac{3}{5}, 0) = (3, 0)$

$$\text{For PQ: } \frac{x_0x}{25} + \frac{y_0y}{16} = 1$$

(x_0, y_0) on D $\therefore x_0 = \frac{25}{3}$ (from (i))

$$\therefore \text{PQ is: } \frac{x}{3} + \frac{y_0y}{16} = 1$$

Substituting $(3, 0)$ gives:

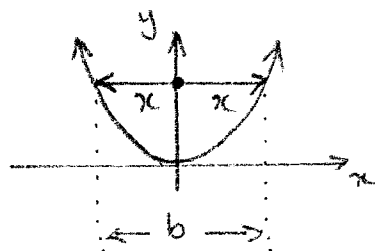
$$\text{LHS} = \frac{3}{3} + 0$$

$$= 1$$

$$= \text{RHS}$$

$\therefore$ Focus lies on PQ (QED) (4)

④ (a) Area of cross-section = $\frac{1}{2}bh$ where $b = h$ for isosceles



where $b = 2x$ and $x = \sqrt{y}$

$$\therefore A = \frac{1}{2} \times 2\sqrt{y} \times 2\sqrt{y} = 2y$$

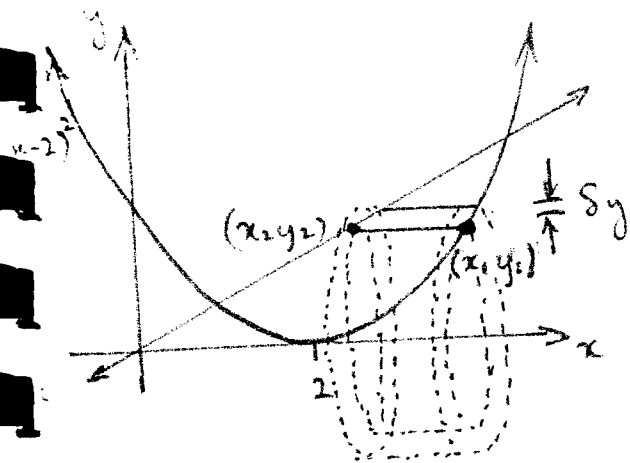
$$\therefore V = \lim_{\delta y \rightarrow 0} \sum_{i=0}^4 2y_i \delta y$$

$$= 2 \int_0^4 y \, dy$$

$$= 2 \left[\frac{1}{2} y^2 \right]_0^4$$

$$= 4^2 - 0^2$$

$$= \boxed{16 \text{ units}^3} \quad (4)$$



boundary values:

$$(x-2)^2 = x$$

$$x^2 - 4x + 4 = x$$

$$x^2 - 5x + 4 = 0$$

$$(x-1)(x-4) = 0$$

$$\therefore x = 1 \text{ or } 4$$

$$y = 1 \text{ or } 4$$

area of annular base

$$= \pi [(y + \delta y)^2 - y^2]$$

$$= 2\pi y \delta y \quad (\delta y^2 \text{ negligible})$$

and $V = Ah$ for shell

$$\therefore V = \lim_{\delta y \rightarrow 0} \sum_1^4 2\pi y \delta y \times x$$

where $x = x_1 - x_2$

$$y_1 = (x_1 - 2)^2 \rightarrow x_1 = \sqrt{y_1} + 2$$

$$y_2 = x_2^2$$

$$\therefore x = \sqrt{y} + 2 - y$$

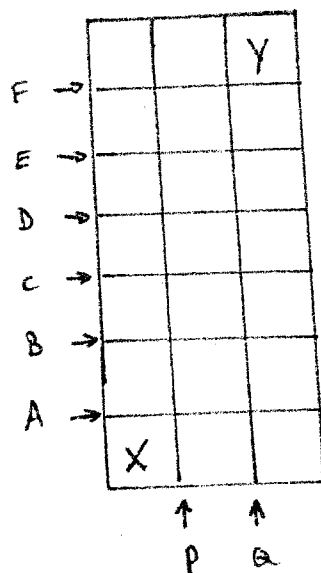
$$\therefore V = 2\pi \int_1^4 y(\sqrt{y} + 2 - y) dy$$

$$= 2\pi \int_1^4 y^{3/2} + 2y - y^2 dy$$

$$= 2\pi \left[\frac{2}{5} y^{5/2} + y^2 - \frac{1}{3} y^3 \right]_1^4$$

$$V = \frac{64\pi}{5} \text{ units}^3$$

(c)(i) Labelling the lines which have to be crossed from X to Y as shown:



and noting that they have to be crossed "in alphabetical order" both horizontally and vertically ...
THEN placing P first:

$$\begin{array}{c} P \\ \hline \end{array} = 7$$

Q in any of 7 possible positions

(the other 6 positions only possible in ONE way i.e. alphabetical: A B C D E F)

$$\text{or } \begin{array}{c} P \\ \hline \end{array} = 6$$

Q in 6 positions

$$\text{or } \begin{array}{c} P \\ \hline \end{array} = 1$$

Q in 1 position

$$\therefore \text{Number of ways} = 7 + 6 + 5 + 4 + \dots + 1 = 28$$

(3)

(6)

(OR) All 8 letter "words" from
 $A B C D E F P Q = 8!$

but within these arrangements
 each 6! arrangements (A to F)
 can only occur ONCE (alphabetical)
 and 2! (P, Q) arrangements
 can only occur ONCE

$$\therefore \text{Total ways} = \frac{8!}{6! \cdot 2!}$$

$$= \textcircled{28}$$

(ii) EITHER

X to M: P -----

P in only 5 positions

M to Y: Q --

Q in only 3 positions

$$\therefore \text{Total} = 5 \times 3$$

$$= \textcircled{15}$$

(2)

$$\textcircled{\text{OR}} \text{ Total ways} = \frac{5!}{4!1!} \times \frac{3!}{2!1!}$$

(X → M) (M → Y)

$$= \textcircled{15}$$

(5) a) i)

$$x^3 - x^2 + 4x - 2 = 0$$

satisfied by $x = a^2$
 i.e. $a = x^{\frac{1}{2}}$

$\therefore$ equation required given by:

$$(x^{\frac{1}{2}})^3 - (x^{\frac{1}{2}})^2 + 4(x^{\frac{1}{2}}) - 2 = 0$$

$$\text{or } x\sqrt{x} - x + 4\sqrt{x} - 2 = 0$$

$$\therefore \sqrt{x}(x+4) = x+2$$

$$\therefore x(x^2 + 8x + 16) = x^2 + 4x + 4$$

$$\therefore x^3 + 8x^2 + 16x - x^2 - 4x - 4 = 0$$

$$\text{i.e. } \textcircled{x^3 + 7x^2 + 12x - 4 = 0}$$

(3)

ii) Using:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

then:

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\{\alpha\beta \cdot \alpha\gamma + \alpha\beta \cdot \beta\gamma + \alpha\gamma \cdot \beta\gamma\}$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2(\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2)$$

$$= (\alpha\beta + \alpha\gamma + \beta\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$\text{where } \alpha + \beta + \gamma = -b/a = 1$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = c/a = 4$$

$$\alpha\beta\gamma = -d/a = 2$$

$$\therefore \text{Answer} = 4^2 - 2 \times 2 \times 1$$

$$= \textcircled{12}$$

(3)

$$p(x) = 4x^3 + 4x^2 + x + k$$

$$p'(x) = 12x^2 + 8x + 1$$

$$\text{For } p'(x) = 0: 12x^2 + 8x + 1 = 0$$

$\Downarrow$

$$(6x+1)(2x+1) = 0$$

$$\therefore x = -\frac{1}{6} \text{ or } -\frac{1}{2} \dots\dots (1)$$

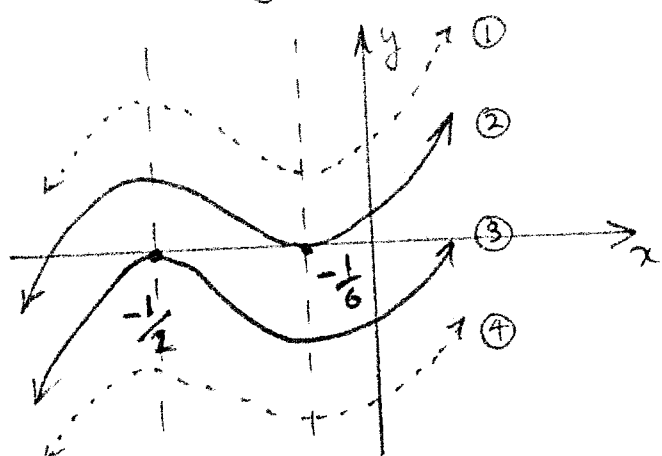
$$\therefore p(-\frac{1}{6}) = 4(-\frac{1}{6})^3 + 4(-\frac{1}{6})^2 + (-\frac{1}{6}) + k$$

$$= -\frac{2}{27} + k \dots\dots (2)$$

$$\text{and } p(-\frac{1}{2}) = 4(-\frac{1}{2})^3 + 4(-\frac{1}{2})^2 + (-\frac{1}{2}) + k$$

$$= k \dots\dots (3)$$

Now, the graph of $y = p(x)$ has two turning points and could be:



$$\text{For } p(-\frac{1}{6}) = 0 \text{ (graph 2)} \quad k = \frac{2}{27} \text{ (from (2))}$$

$$p(-\frac{1}{2}) = 0 \text{ (graph 3)} \quad k = 0 \text{ (from (3))}$$

$\therefore$ for one real root (graphs 1/4)

$$\text{we need } k < 0 \text{ or } k > \frac{2}{27} \quad (4)$$

$$c) p(x) = 3x^4 - 11x^3 + 14x^2 - 11x + 3$$

$$= x^2 \left\{ 3x^2 - 11x + 14 - \frac{11}{x} + \frac{3}{x^2} \right\}$$

$$= x^2 \left\{ \left[3x^2 + 6 + \frac{3}{x^2} \right] + \left[-11x - \frac{11}{x} \right] + 8 \right\}$$

$$= x^2 \left\{ 3 \left[x^2 + 2 + \frac{1}{x^2} \right] - 11 \left[x + \frac{1}{x} \right] + 8 \right\}$$

$$= x^2 \left\{ 3 \left(x + \frac{1}{x} \right)^2 - 11 \left(x + \frac{1}{x} \right) + 8 \right\}$$

(QED)

$$\text{For } p(x) = 0 \quad x^2 \neq 0$$

i.e. $x = 0$ is not a solution

$$\therefore 3 \left(x + \frac{1}{x} \right)^2 - 11 \left(x + \frac{1}{x} \right) + 8 = 0$$

$$\text{Letting } A = x + \frac{1}{x}$$

$$\therefore 3A^2 - 11A + 8 = 0$$

$$(3A-8)(A-1) = 0$$

$$\therefore 3 \left(x + \frac{1}{x} \right) - 8 = 0 \text{ or } \left(x + \frac{1}{x} \right) - 1 = 0$$

$$\therefore 3x^2 - 8x + 3 = 0$$

$$x^2 - x + 1 = 0$$

$$\therefore x = \frac{8 \pm \sqrt{28}}{6}$$

$$x = \frac{1 \pm \sqrt{-3}}{2}$$

$$\therefore \text{For } p(x) = 0$$

$$x = \frac{4 \pm \sqrt{7}}{3}, \quad \frac{1 \pm i\sqrt{3}}{2} \text{ (over } \mathbb{C} \text{)}$$

$$\text{and } p(x) = \left(x - \frac{4 - \sqrt{7}}{3} \right) \left(x - \frac{4 + \sqrt{7}}{3} \right) (x^2 - x + 1)$$

(over $\mathbb{R}$)

a) i) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $P(a \sec \theta, b \tan \theta)$

Equation of normal:

$$y - y_1 = m(x - x_1)$$

for m: $\frac{2x}{a^2} - \frac{2y}{b^2} \cdot y' = 1$

$$\therefore y' = \frac{b^2 x}{a^2 y}$$

$$\therefore \text{at } P, m = \left. \begin{aligned} &\frac{b^2 \cdot a \sec \theta}{a^2 \cdot b \tan \theta} \\ &= \frac{b}{a \sin \theta} \end{aligned} \right\} \text{TANGENT}$$

$$\therefore m = -\frac{a \sin \theta}{b} \left\} \text{NORMAL}$$

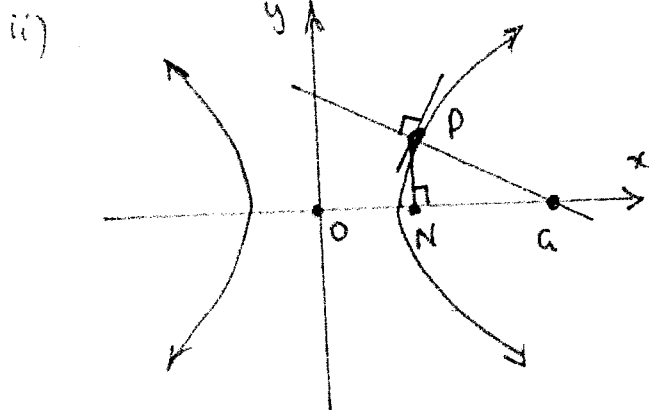
$\therefore$ Equation of normal:

$$y - b \tan \theta = -\frac{a \sin \theta}{b} (x - a \sec \theta)$$

$$\therefore by - b^2 \tan \theta = -a \sin \theta x + a^2 \tan \theta$$

$$\therefore a \sin \theta x + by = (a^2 + b^2) \tan \theta \quad (\text{QED})$$

(4)



At N: $x = x_P = a \sec \theta$

$$\therefore ON = a \sec \theta \quad \dots \quad (1)$$

At Q: let $y = 0$

$$\therefore a \sin \theta x = (a^2 + b^2) \tan \theta$$

$$\therefore x = \frac{a^2 + b^2}{a} \sec \theta$$

$$\therefore OQ = \frac{a^2 + b^2}{a} \sec \theta$$

$$\text{and } b^2 = a^2(e^2 - 1) \rightarrow e^2 = 1 + \frac{b^2}{a^2} = \frac{a^2 + b^2}{a^2}$$

$$\therefore OQ = a \times \frac{a^2 + b^2}{a^2} \sec \theta$$

$$= a \times e^2 \sec \theta$$

$$= e^2 a \sec \theta$$

$$\therefore OQ = e^2 \times ON \quad (\text{QED}) \quad (5)$$

b) In $\triangle OKL$: $\angle LOK = \angle OLK$ (isos.)

$$\therefore \angle LOK = \frac{\pi}{6} \quad (\angle \text{sum of } \triangle)$$

In $\triangle OML$: $\angle LOM = \frac{\pi}{3}$ (equil $\triangle$)

$$\therefore \angle KOM = \frac{\pi}{2}$$

i.e. $\beta \equiv \alpha$ rotated $90^\circ \equiv$ (X by i) ... (1)
(OM) (OK)

AND In $\triangle OKL$:

$$\cos \angle LOK = \frac{\frac{1}{2} OL}{OK} = \frac{\frac{1}{2} OL}{\alpha}$$

$$\text{i.e. } \cos \frac{\pi}{6} = \frac{\frac{1}{2} OL}{\alpha} = \frac{\sqrt{3}}{2}$$

$$\therefore OL = \sqrt{3} \alpha$$

In $\triangle OML$: $OM = OL$

$$\therefore \beta = \sqrt{3} \alpha \quad (\text{in magnitude}) \quad \dots \quad (2)$$

from (1) / (2): $\sqrt{3} \alpha = i \beta$ (square both)

$$3\alpha^2 = -\beta^2$$

$$\therefore 3\alpha^2 + \beta^2 = 0 \quad (\text{QED})$$

a) Prove, by induction:

$$\cos \frac{90^\circ}{2^n} = \frac{1}{2} \sqrt{2 + \underbrace{\sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}_{n \text{ terms}}}$$

prove true for $n=1$:

$$\begin{aligned} \text{LHS} &= \cos \frac{90^\circ}{2^1} \\ &= \cos 45^\circ \\ &= \frac{1}{\sqrt{2}} \quad (\text{exact L}) \end{aligned} \quad \begin{aligned} \text{RHS} &= \frac{1}{2} \sqrt{2} \\ &\quad \uparrow \\ &\quad 1 \text{ term} \\ &= \frac{\sqrt{2} \times \sqrt{2}}{2 \times \sqrt{2}} \\ &= \frac{2}{2\sqrt{2}} \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$:

$$\cos \frac{90^\circ}{2^k} = \frac{1}{2} \sqrt{2 + \underbrace{\sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}_{k \text{ terms}}}$$

(here: we see 3 terms)

to prove true for $n=k+1$:

we want:

$$\cos \frac{90^\circ}{2^{k+1}} = \frac{1}{2} \sqrt{2 + \underbrace{\sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}}_{k+1 \text{ terms}}}$$

(here: we want to see 4 terms)

$$\text{LHS: Let } \cos \frac{90^\circ}{2^k \times 2} = \cos \theta$$

$$\therefore \cos 2\theta = \cos \frac{90^\circ}{2^k} = 2 \cos^2 \theta - 1$$

$$\begin{aligned} \therefore \cos^2 \theta &= \frac{1}{2}(1 + \cos 2\theta) \\ &= \frac{1}{2}(1 + \cos \frac{90^\circ}{2^k}) \end{aligned}$$

$$= \frac{1}{2} \left(1 + \frac{1}{2} \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \right) \text{ from assumption}$$

..... (1)

$$\begin{aligned} \therefore \text{LHS} &= \cos \theta \\ &= \sqrt{(1)} \end{aligned}$$

$$\begin{aligned} &= \frac{1}{\sqrt{2}} \sqrt{\left(1 + \frac{1}{2} \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \right)} \\ &= \frac{1}{\sqrt{2}} \sqrt{\frac{1}{2} (2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}})} \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \\ &= \frac{1}{2} \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} \\ &\quad \underbrace{\hspace{10em}}_{k+1 \text{ (or 4) terms}} \end{aligned}$$

$$= \text{RHS}$$

$$\text{i.e. LHS} = \text{RHS}$$

$\therefore$ true for $n=k+1$ when true

for $n=k$ and true for $n=1$

$\therefore$ true for $n=2, 3, 4$ i.e. $n \geq 1$

i.e. true, by maths induction (5) (QED).

2) (i) Prove $\tan^{-1}(n+1) - \tan^{-1}(n) = \cot^{-1}(1+n+n^2)$

Let $\tan^{-1}(n+1) = \alpha$

$\therefore \tan \alpha = n+1$

and $\tan^{-1}(n) = \beta$

$\therefore \tan \beta = n$

Now: LHS = $\alpha - \beta$

and $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

$= \frac{n+1 - n}{1 + (n+1)n}$

$= \frac{1}{1+n+n^2}$

$\therefore \cot(\alpha - \beta) = 1+n+n^2$

$\therefore \alpha - \beta = \cot^{-1}(1+n+n^2)$
= RHS

i.e. LHS = RHS (QED) (3)

(ii) $\cot^{-1} 3 = \cot^{-1}(1+1+1^2)$
 $= \tan^{-1} 2 - \tan^{-1} 1$ (from (i))

$\cot^{-1} 7 = \cot^{-1}(1+2+2^2)$
 $= \tan^{-1} 3 - \tan^{-1} 2$

$\cot^{-1} 13 = \cot^{-1}(1+3+3^2)$
 $= \tan^{-1} 4 - \tan^{-1} 3$

$\therefore \cot^{-1} 3 + \cot^{-1} 7 + \cot^{-1} 13 + \dots$
 $+ \cot^{-1}(1+n+n^2)$

$= \tan^{-1} 2 - \tan^{-1} 1 + \tan^{-1} 3 - \tan^{-1} 2$
 $+ \tan^{-1} 4 - \tan^{-1} 3 + \dots$
 $+ \tan^{-1}(n+1) - \tan^{-1}(n)$

giving:

$\tan^{-1}(n+1) - \tan^{-1} 1$

or $\tan^{-1}(n+1) - \frac{\pi}{4}$ (3)

c) Sketch:

• $y_1 = \frac{1}{2} + \sin x$ i.e. $\sin x$ raised $\frac{1}{2}$.

• $y_2 = (\frac{1}{2} + \sin x)^3 = (y_1)^3$

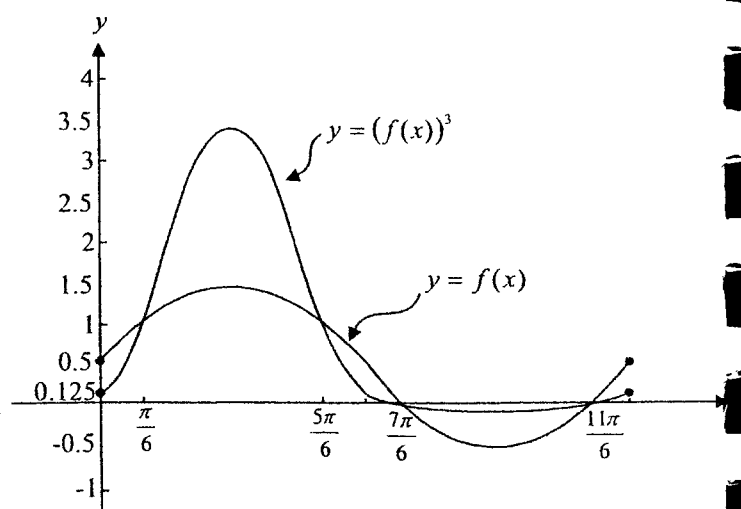
i.e. $(y_1)^3$:

critical points
for $f(x) > (f(x))^3$

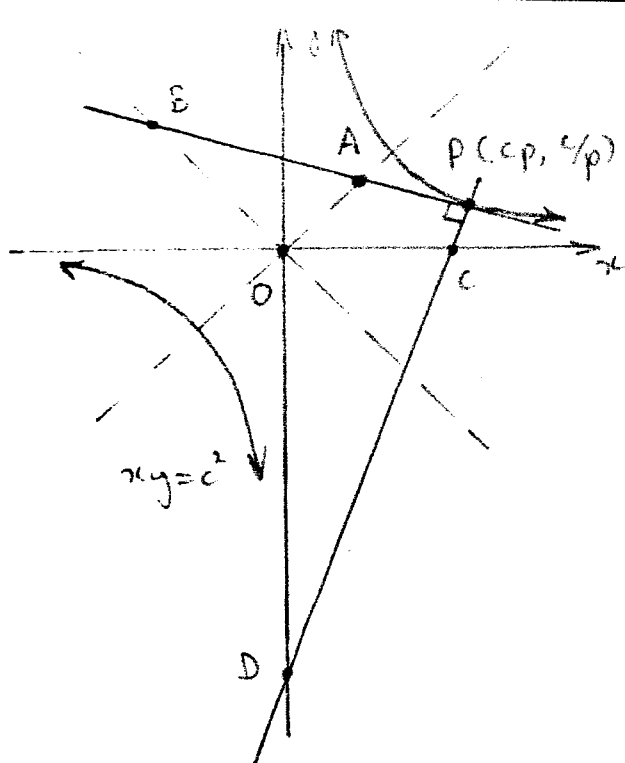
y_1	$(y_1)^3$
0.5	0.125
1	1 (cross over)
1.5	3.375
0	0 (x-int's)
-0.5	-0.125

see graph

(*) $\frac{1}{2} + \sin x = 1 \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$



clearly: $f(x)$ "above" $(f(x))^3$ for
 $0 < x < \frac{\pi}{6}$, $\frac{5\pi}{6} < x < \frac{7\pi}{6}$, $\frac{11\pi}{6} < x < 2\pi$



Equations of Tangent / Normal

$$y - y_1 = m(x - x_1)$$

for m: $xy = c^2$ } implicit
 $\therefore xy' + y = 0$ } diff'n

$$\therefore y' = -y/x$$

$$\therefore \text{at } P: m_T = \frac{-c/p}{cp} = -1/p^2$$

$$\therefore m_N = p^2$$

Tangent: $y - c/p = -1/p^2(x - cp)$

$$\therefore x + p^2y - 2pc = 0 \dots (1)$$

Normal: $y - c/p = p^2(x - cp)$

$$\therefore p^3x - py - c(p^4 - 1) = 0 \dots (2)$$

At A: $y = x$ in (1)

$$\therefore x + p^2x - 2pc = 0$$

$$\therefore A \text{ is: } \left(\frac{2pc}{1+p^2}, \frac{2pc}{1+p^2} \right) \dots (3)$$

Similarly, at B, $y = -x$

$$\therefore B \text{ is: } \left(\frac{2pc}{1-p^2}, \frac{2pc}{1-p^2} \right) \dots (4)$$

At C: $y = 0$

$$\therefore p^3x = c(p^4 - 1)$$

$$x = \frac{c(p^4 - 1)}{p^3} \Rightarrow OC$$

At D: $x = 0$

$$\therefore py = -c(p^4 - 1)$$

$$y = \frac{-c(p^4 - 1)}{p}$$

(ie $OD = \frac{c(p^4 - 1)}{p}$: for distance)

from (3)/(4) 45° right Δ's:

give: $OA = \frac{2\sqrt{2}pc}{1+p^2}$, $OB = \frac{2\sqrt{2}pc}{1-p^2}$

$$\therefore M^2N = \left(\frac{1}{2} \times \frac{2\sqrt{2}pc}{1+p^2} \times \frac{2\sqrt{2}pc}{1-p^2} \right)^2$$

$$\times \left(\frac{1}{2} \times \frac{c(p^4 - 1)}{p^3} \times \frac{c(p^4 - 1)}{p} \right)$$

$$= \left(\frac{4p^2c^2}{1-p^4} \right)^2 \times \frac{c^2(p^4 - 1)^2}{2p^4}$$

$$= \frac{16p^4c^4}{(1-p^4)^2} \times \frac{c^2(p^4 - 1)^2}{2p^4}$$

$$= 8c^2$$

ie. M^2N is constant (QED)
 (for constant c).

$$a) f(x) = \frac{1-|x|}{|x|}$$

$$i) f(a) = \frac{1-|a|}{|a|}$$

$$f(-a) = \frac{1-|-a|}{|-a|}$$

$$= \frac{1-|a|}{|a|}$$

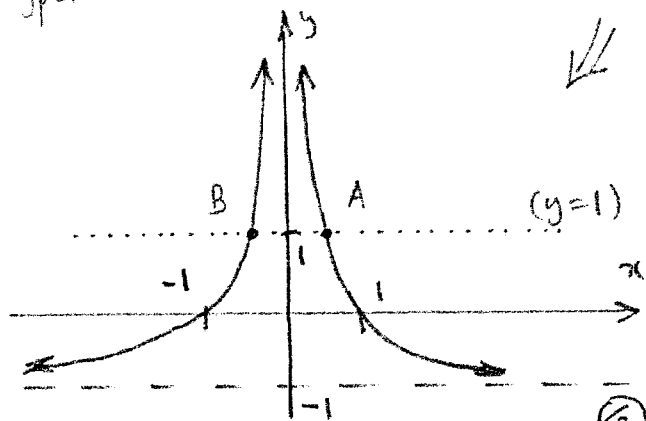
$$\therefore f(a) = f(-a) \therefore \text{EVEN} \quad (1)$$

$$(ii) f(x) = \frac{1-|x|}{|x|}$$

$$= \frac{1}{|x|} - \frac{|x|}{|x|}$$

$$= \frac{1}{|x|} - 1 \quad \left(\frac{1}{|x|} = \left| \frac{1}{x} \right| \right)$$

hyperbola reflected above x-axis



(iii) For $f(x) \geq 1$, find A/B

A: intersection with $\frac{1}{x} - 1$ and 1

$$\text{i.e. } \frac{1}{x} = 2$$

$$x = \frac{1}{2}$$

B: intersection with $-\frac{1}{x} - 1$ and 1

$$\text{i.e. } -\frac{1}{x} = 2$$

$$\therefore x = -\frac{1}{2}$$

and $y = f(x)$ ABOVE/on $y=1$
($\geq$)

for

$$\boxed{-\frac{1}{2} \leq x \leq \frac{1}{2}}$$

(iv) • For $x < -1$ } $-1 < f(x) < 0$
 $x > 1$ }

$$\therefore e^{-1} < e^{f(x)} < e^0$$

$$\frac{1}{e} < e^{f(x)} < 1$$

• For $x = \pm 1$: $f(x) = 0$

$$\therefore e^{f(x)} = 1$$

• For $-1 < x < 1$: $f(x) \rightarrow \infty$

$$\therefore e^{f(x)} \rightarrow \infty$$

